

CHALMERS	Anonymous code		Consecutive page no Löpande sid nr
	Anonym kod		Question no. Uppgift nr 1

Basic question 1: Hash tables

Insert the following values into initially empty hash tables, in the specified order:

Axe, Bo, Center, D, E, Five, Graph

Use the following (poor-quality) hash function, with modular compression:

$h(x) = \text{length of } x$ (remember to double-check the long words!)

You do not have to resize the hash tables.

(a) Use a linear probing hash table with capacity 10.

0	
1	
2	
3	
4	
5	
6	
7	
8	
9	

(b) Use a separate chaining hash table with capacity 5, and standard linked lists.

0	
1	
2	
3	
4	

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Basic question 2: Heaps

Here is a binary min-heap of integers in array form, with an unknown value x :

0	1	2	3	4	5	6	7	8	9
2	5	2	x	8	10	3	12	7	8

List all the possible values for x :

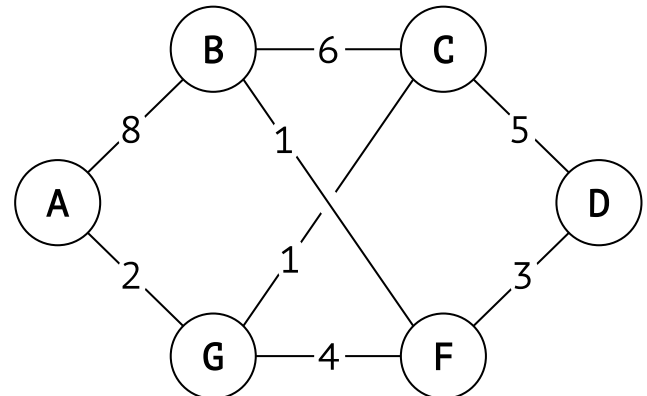
Explain why this is the case.

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Basic question 3: Graphs

Construct (a) a minimum spanning tree (MST), and (b) a shortest path tree (SPT) rooted at **D**, for the weighted undirected graph on the right.

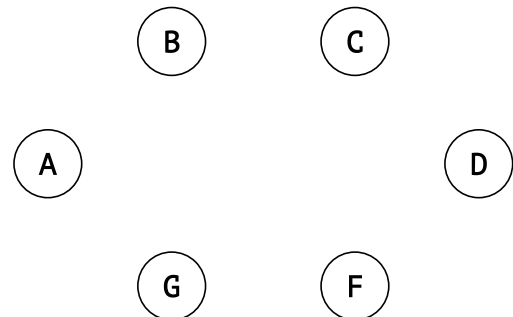
For (a) you should run Prim's algorithm from **A**, and for (b) you should run Dijkstra's from **D**.



(a) *Prim's algorithm starting in A*: How does the MST look? (Draw the edges.)

List the vertices in the order they are visited by Prim's algorithm:

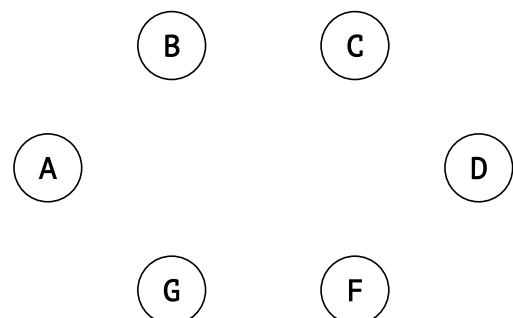
A, _____



(b) *Dijkstra's algorithm starting in D*: How does the SPT look? (Draw the edges.)

List the vertices in the order they are visited by Dijkstra's algorithm:

D, _____



Basic question 4: Insertion and selection sort

Perform in-place insertion sort *and* in-place selection sort on the array [81, 63, 32, 75, 59, 41].
Show the array after each iteration of the outer loop in the algorithm.

Insertion sort

0	1	2	3	4	5
81	63	32	75	59	41

0	1	2	3	4	5

0	1	2	3	4	5

0	1	2	3	4	5

0	1	2	3	4	5

0	1	2	3	4	5

Selection sort

0	1	2	3	4	5
81	63	32	75	59	41

0	1	2	3	4	5

0	1	2	3	4	5

0	1	2	3	4	5

0	1	2	3	4	5

0	1	2	3	4	5

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Basic question 5: Complexity

The following function takes an array of **distinct** positive integers (that is, no duplicates) and produces a new larger array with the same elements and some cells that are null, much as resizing works in a linear probing hash table.

`build_table(arr: integer array) → (integer or null) array:`

`newLength = arr.length · 2`

`newArr = new array of length newLength, all cells null`

for `x` **in** `arr`:

`ix = x mod newLength`

while `newArr[ix]` **is not** null:

`ix = (ix + 1) mod newLength`

`newArr[ix] = x`

return `newArr`

- a) What is the asymptotic **worst-case** complexity of `build_table(arr)` in terms of the length n of `arr`? Justify your answers with comments in the margin above.

Complexity: _____

- b) Construct a worst-case input of size 5 (an array of length 5) for which the algorithm is as slow as possible. Keep in mind that there can be no duplicate values!

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Basic question 6: AVL size imbalance

Recall that AVL trees satisfy a height balance invariant: *the heights of siblings differ by at most 1*.

However, the **sizes** of siblings can differ more. Draw an AVL tree with 10 nodes, containing the numbers 0 to 9, such that the sizes of the two children of the root differ as much as possible.

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Advanced question 7: Remove from a strange tree

A *strange tree* is a data structure storing a set of values. It is a binary tree, but not a BST.

A strange tree is either *null* or has a value and left and right children satisfying the following:

- The value is *less than* all values in its subtrees (that is, a strict version of the heap property holds).
- All the values in the left subtree are *less than* all the values in the right subtree.
- Both children are also strange trees.
- An example strange tree is to the right (*null* pointers are omitted).

You may assume that the StrangeTree class looks like this:

```
class StrangeTree:
    root: TreeNode

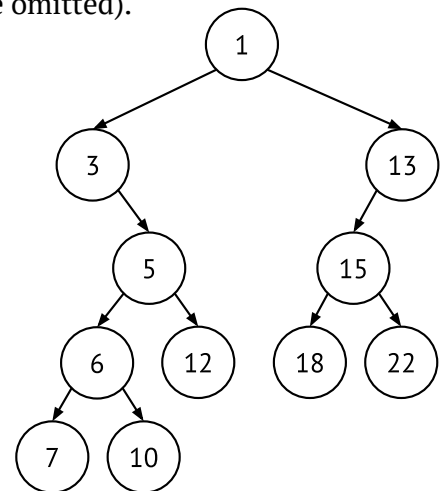
class TreeNode:
    value: int
    left, right: TreeNode
```

Your task is to write two functions that remove the minimal and maximum values from a strange tree, respectively:

- removeMin(tree: StrangeTree)
- removeMax(tree: StrangeTree)

Notes:

- You can assume that the input tree is not *null*.
- The resulting tree must still be a strange tree, of course.
- Your solution should take $O(h)$ time in terms of the height h of the tree.
- You may answer using pseudocode or Java/Python/Haskell.
- You can define helper functions if you want.



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Advanced question 8: Complexity

You have a database of the N most prominent European rivers. The database consists of a number of sorted arrays of length N , including the following:

- **rivers** stores the names of the rivers, in alphabetical order
- **lengths** stores the length of each river, in *the same order* as the **rivers** array
- **length_index** contains the river names, but sorted by *the length of the river*
- (there can be more arrays storing the basin size, or the average discharge, or other things for each river – but we ignore them here)

Here is an example of how the database can look:

rivers		lengths		length_index	
0	Adige	0	410 km	0	Yantra
1	Belaya	1	1430 km	1	Thames
2	Dalälven	2	520 km	2	Adige
3	Danube	3	2860 km	3	Dalälven
...		...		...	
$N-2$	Volga	$N-2$	3690 km	$N-2$	Danube
$N-1$	Yantra	$N-1$	285 km	$N-1$	Volga

Consider the following problem:

“Given a length K kilometers, which is the longest river that is shorter than K ?”

We want to solve this problem using a fast algorithm. What is the best asymptotic complexity in N you can achieve? Write your answer in O -notation. Justify by briefly sketching your algorithm.

- You can assume that number and string comparisons take constant time.
- You do not have to justify why there is no better algorithm.

Complexity: _____

Justification (algorithm sketch):

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Advanced question 9: Mystery data structure

Below is pseudo-code for a data structure from the course.

- What data structure is it? Be specific.
- What is a better name for the Mystery variable **b**?
- For each of the operations **foo**, **bar** and **baz** on the data structure, state briefly (in one or two sentences) what they do and their asymptotic (worst-case) complexity. Also state the asymptotic *amortised* (worst-case) complexity if it is different.

data Mystery:

a: array of values

b: integer

function foo(m: Mystery):

i = 0

repeat:

g = i · 2 + 1

if not g < m.b:

return

if g + 1 < m.b **and** m.a[g+1] > m.a[g]:

g = g + 1

if m.a[g] ≤ m.a[i]:

return

tmp = m.a[i]

m.a[i] = m.a[g]

m.a[g] = tmp

i = g

function bar(m: Mystery) → value:

result = m.a[0]

m.a[0] = m.a[m.b-1]

m.a[m.b-1] = null

m.b = m.b - 1

if m.b < m.a.length / 4:

arr = **new array of length** m.b · 2

for i in 0 ... m.b-1:

arr[i] = m.a[i]

m.a = arr

foo(m)

return result

function baz(m: Mystery) → array of values:

arr = **new array of length** m.b

for i in 0 ... m.b-1:

arr[i] = bar(m)

return arr